Hypertension, Preterm labor, Premature rupture of membranes, Low birth weight (<2,500 g), Operative delivery, Fetal death and/or Cerebral palsy. (Mullin, 2010)

Even though eSET requires subsequent frozen embryo transfer cycle(s) if the initial fresh cycle is unsuccessful, it is prudent to promote elective single blastocyst embryo transfer as a means of reducing the frequency of multiple gestations and the associated risks of poor maternal and birth outcomes. (Johnson, 2013; Sunderam, 2012). Numerous countries have adopted regulations that mandate eSET resulting in a twin gestation rate of <5%. Pregnancy rates for eSET are comparable to multiple embryo transfer. (Thurin, 2004). Although pregnancy outcome diminishes with increasing maternal age, all age groups should be considered for blastocyst stage eSET (Niinimaki, 2012; Kato, 2012) particularly in the context of preimplantation genetic testing or other technologies that enhance the embryo selection process.

Strategies to improve live birth have primarily focused on maximizing embryo selection and endometrial synchrony. These strategies include PGT-A, freezing only embryo transfer cycles, endometrial synchrony testing, and time lapse imaging and other noninvasive embryo testing strategies. There is currently a lack of robust and consistent evidence that these strategies improve the chances of achieving a live birth. In addition, while strategies to improve live birth are aimed at improving the live birth success of each embryo transfer, they are not necessary to reduce multiple gestation. Performing SET without additional embryo or endometrial testing reduces the multiple gestation rate down to the background 1–2% risk of monozygotic twins in ART (Gardner 2004, Styer 2008, Crinit 2005). Single embryo transfer, regardless of additional testing, should be considered the gold standard to reduce multiple gestation. Given the lack of robust evidence or conflicting evidence for many of these tests to improve clinical outcomes, they are currently not routinely recommended as a strategy to increase SET (ASRM 2022)

Embryo Banking and Use of Frozen Embryos

There is no evidence in medical literature to support the practice of repeated ART cycles for the purpose of accumulating (banking) embryos for later use (egg retrievals without a fresh or frozen embryo transfer) with the exception of freeze all cycles for medical necessity. It is clinically appropriate and cost effective to utilize all frozen embryos for transfer prior to another fresh ART cycle. (Forman, 2013; Richter, 2006; Shapiro, 2011, 2013).

Cryopreservation

Cryopreservation is freezing at a very low temperature, such as in liquid nitrogen (-196C) to keep embryos, eggs, or sperm viable. Sub-zero temperature is not a physiological condition for cells and water ice crystals represent the main problem since they induce cell death, principally in large cells like oocytes, which have a meiotic spindle that degenerates during this process. Significantly, cryopreservation represents an option for fertility preservation in patients who develop gonadal failure for any condition and those who want to freeze their germ cells for later use (Estudillo, et. al., 2021).